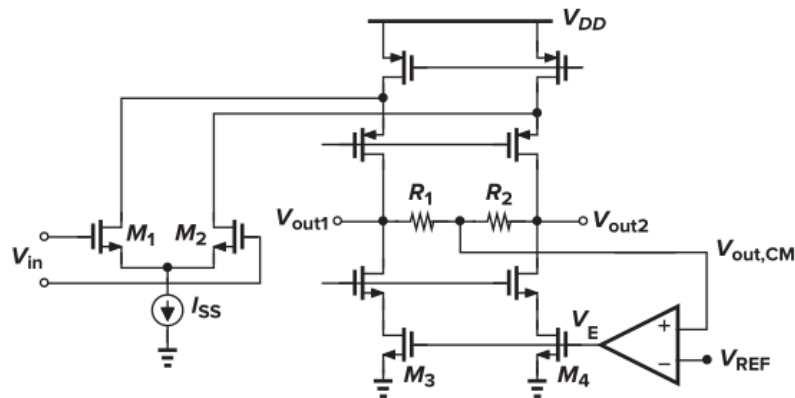
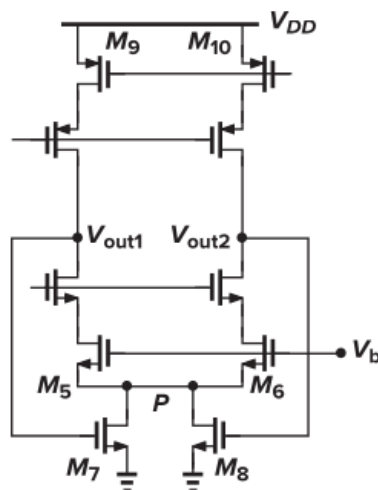


Ex 1: Consider the CMFB network in figure below. The amplifier sensing $V_{out,CM}$ is to be implemented as a differential pair with active current mirror load.

- Should the input pair of the amplifier use PMOS devices or NMOS devices?
- Calculate the loop gain for the CMFB network.



Ex 2: In the circuit of figure below, each branch is biased at a current of 0.5 mA. Choose the dimensions of M_7 and M_8 such that the output CM level is equal to 1.5 V and $V_P = 100$ mV.



Ex 3: Consider the transimpedance amplifier shown in figure below, where $R_D = 1 \text{ k}\Omega$, $R_F = 10 \text{ k}\Omega$, $g_{m1} = g_{m2} = 1/(100\Omega)$, and $C_A = C_X = C_Y = 100 \text{ fF}$. Neglecting all other capacitances and assuming that $\lambda = \gamma = 0$, compute the phase margin of the circuit.

